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Analysis of kinematic factors influencing performance in the Chinese men's 4 × 100 m relay team

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To investigate the key factors influencing performance in the men's 4 × 100 m relay, this study analyzed the relationship between baton exchange kinematics and relay zone performance. High-speed cameras were employed to record the six national-level sprinters' movements and velocities. Relay zone 30 m time showed statistically significant correlations with the velocity of the delivery leg when entering the relay zone ($r = -0.600$, $P < 0.01$), the velocity of the receiving leg at the start of the baton-handover ($r = -0.502$, $P < 0.01$), the velocity of the receiving leg at 10 m and 15 m ($r = -0.411$, $P < 0.05$; $r = -0.474$, $P < 0.01$, respectively), and handover time ($r = 0.389$, $P < 0.05$). Stepwise regression analysis revealed that the velocity of the delivery leg when entering the relay zone ($P < 0.001$) and the velocity of the receiving leg at the start of the baton-handover ($P = 0.002$) collectively accounted for 55% of the variance in relay zone 30 m time. Therefore, targeted training should therefore prioritize maximizing the velocity of the delivery leg when entering the relay zone, improving the velocity of the delivery leg when handover, and minimizing handover time to enhance relay zone efficiency and elite-level relay performance.

Keywords Sprinting, 4 × 100m, Relay, Velocity, Technique

The 4 × 100 m relay, a premier team speed racing event, involves four athletes who must cover the designated distance while sprinting at maximum velocity to execute the baton handover within the defined relay zone[1]. Historically, it has not consistently been true that teams exhibiting superior speed achieve better performance outcomes. For instance, during the 2024 Paris Olympic 4 × 100 m relay final, the United States team, despite fielding four of the fastest athletes, faced disqualification due to a dropped baton. Conversely, the Canadian team, which secured first place, comprised athletes whose seasonal bests in the 100 m were 0.46 s and 0.34 s slower than those of the second-place South African team and the third-place Italian team, respectively, yet they emerged victorious. Prior research indicates that an adept baton handoff can partially mitigate deficiencies in speed; however, the act of dropping the baton significantly undermines the prospects for victory[2–4].

The performance within the relay zone is influenced by various factors pertaining to both the delivery leg and receiving leg, including baton exchange technique, timing of the receiving leg's start, free distance, handover position, and duration, among others[1,4–9]. In the context of baton exchanges, the "push-forward pass" offers the benefits of the "up sweep-pass" without the associated risk of dropping the baton, thereby maximizing the free distance gained. Conversely, the speed loss incurred with the "up sweep-pass" is less pronounced than that associated with the "down-sweep pass" and "push-forward pass"[10–12]. Factors such as the starting position of the receiving leg, lane assignment, free distance, and handoff position play a crucial role, with minor adjustments in these elements potentially resulting in significant variations in overall performance[5,13]. In an efficient exchange, the baton transfer occurs over two strides in less than one second, with an optimal free distance of 1.5 to 2 m. Furthermore, the effectiveness of the handover—characterized by a shorter handover time—improves as the baton exchange occurs further along in the relay zone, a trend that is particularly pronounced among female athletes[4,14,15]. A case study involving the Japanese men's national relay team, which was preparing for the Tokyo Olympics, demonstrated that the mean velocity of the receiving leg over the 10-meter section from the takeover zone exit was significantly correlated with 40 m performance. Additionally, both baton position and free distance exhibited significant correlations with 20 m performance[8]. This study offers a methodological foundation and practical rationale for the techniques employed in the sprint 4 × 100 m relay.

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In addition to technical factors, an effective handover necessitates that both the delivery leg and the receiving leg attain the highest possible speeds to facilitate the baton exchange. It is evident that the delivery leg experiences a gradual decrease in speed due to the approximately 100-meter run, while the receiving leg exhibits an increase in speed within the relay zone. Consequently, the delivery leg should sustain maximal speed, while the receiving leg must accelerate to its maximum velocity[4,16]. However, empirical research on the relationship between the delivery and receiving leg velocities and relay zone performance remains limited, constraining the scientific guidance available for optimizing relay pacing strategies. In particular, since the International Association of Athletics Federations (IAAF) extended the relay zone length from 20 m to 30 m in 2017, systematic investigations into the key technical factors affecting performance and the velocity patterns of both legs within the 30 m relay zone under the new regulations have been lacking. Therefore, conducting such research not only helps to fill this theoretical gap but also provides coaches and athletes with quantifiable evidence for relay training and tactical decision-making.

Consequently, the primary objective of this research was to elucidate the impact of handover technique and the speeds of both the delivery leg and receiving leg on relay zone performance. This information aims to establish a theoretical and technical foundation for the training of 4 × 100 m relay team athletes, thereby enabling coaches to gain a clearer understanding of the essential characteristics of this event. It is hypothesized that the athletes' speed capabilities and the technical execution of baton handover are critical factors influencing performance within the relay zone.

Methods

Participants

A total of six elite sprinters were meticulously selected based on their performance at the National Athletics Championships. These athletes have engaged in a rigorous training regimen for a duration of two to three months in preparation for the upcoming Hangzhou Asian Games and the Paris Olympic Games. All selected sprinters are of international caliber, and their fundamental characteristics are summarized in Table 1. All participants signed informed consent forms prior to the study, which included authorization for the use of identifiable images. The study was approved by the ethics committee of the local university (No. 2024109 H). All methods were carried out in accordance with the Declaration of Helsinki and relevant regulations.

Procedures and measurements

In the lead-up to the 2023 Hangzhou Asian Games and the 2024 Paris Olympic Games, the Chinese men's 4 × 100 m relay team engaged in a rigorous training program. This study involved the recording of video footage for 29 critical baton exchange training sessions within the relay zones throughout the intensive training period.

Athletes engaged in baton exchange training within each of the three relay zones, initiating the delivery leg's run from a distance of 80 m prior to the relay zone, mirroring the conditions of an actual relay competition. The receiving leg utilized markers placed 30 to 35 feet in advance of the relay zone to establish the timing for the start of the sprint. Within the relay zone, athletes executed the baton handover, with the receiving leg sprinting at full speed to a distance of 10 m behind the relay zone. As illustrated in Fig. 1, the three relay zones were recorded using two high-speed cameras: the primary camera, a Z CAM (Lumix G VARIO F3.5-5.6II, Panasonic, 239.76fps), employed a planar fixed-point tracking method positioned centrally within the bend, featuring a framing range of 15 m and a shooting height of 1.2 m, while a second camera (Lumix G VARIO F3.5-5.6II, Panasonic, 239.76fps) was positioned alongside the straight section, with a framing range of 20 m and a shooting height of 1.5 m. To more effectively capture the athletes' speed during the initial 80 m of the second relay zone, a Z CAM camera (Lumix G VARIO F3.5-5.6II, Panasonic, 239.76fps) was strategically positioned at the 50-meter mark across the center of the straight section, featuring a framing range of 15 m and a shooting height of 1.2 m for planar spot tracking shots. To enhance the accuracy of athlete positioning, a single marker was placed in both the inner and outer lanes, 80 m prior to the relay zone, with additional markers positioned every 5 m both inside and outside the relay zone extending 20 m in front, 30 m within, and 10 m behind the relay zone.

Video analysis and data processing

The motion analysis software, Kinovea (version 2023.1.2; https://www.kinovea.org/), was employed to conduct a comprehensive analysis of the video footage. The variables selected to characterize performance within each relay zone encompassed split time, handover position, handover time, and the free distance of the athletes. Specifically, the split time is defined by the athlete's torso and the inner and outer track markings in a vertical cross-section as a reference, measured in seconds (s); relay zone performance refers to the duration from the moment the delivery leg enters the 30 m relay zone until the receiving leg exits, also measured in seconds (s); the handover position denotes the point at which the delivery leg transfers the baton to the initial contact of the receiving leg, measured in meters (m); and the handover time is the interval between the first and last contact of the baton between the delivery and receiving legs, again measured in seconds (s). The free distance is defined as the spatial separation between the two legs at the moment the delivery leg transfers the baton to the initial contact of the receiving leg, measured in meters (m), as illustrated in Fig. 2. The athlete's velocity was subjected to linear fitting in relation to the split time recorded at every 5-meter interval. The sample interpolation rate was

Variables	Ages (y)	Height (m)	Body mass (kg)	SB (s)	PB (s)
Mean ± SD	25 ± 4.38	1.81 ± 0.03	73.33 ± 8.14	10.18 ± 0.08	10.13 ± 0.10

Table 1. Basic information of men's 4 × 100 m athletes. *SB* season's best, *PB* personal best.

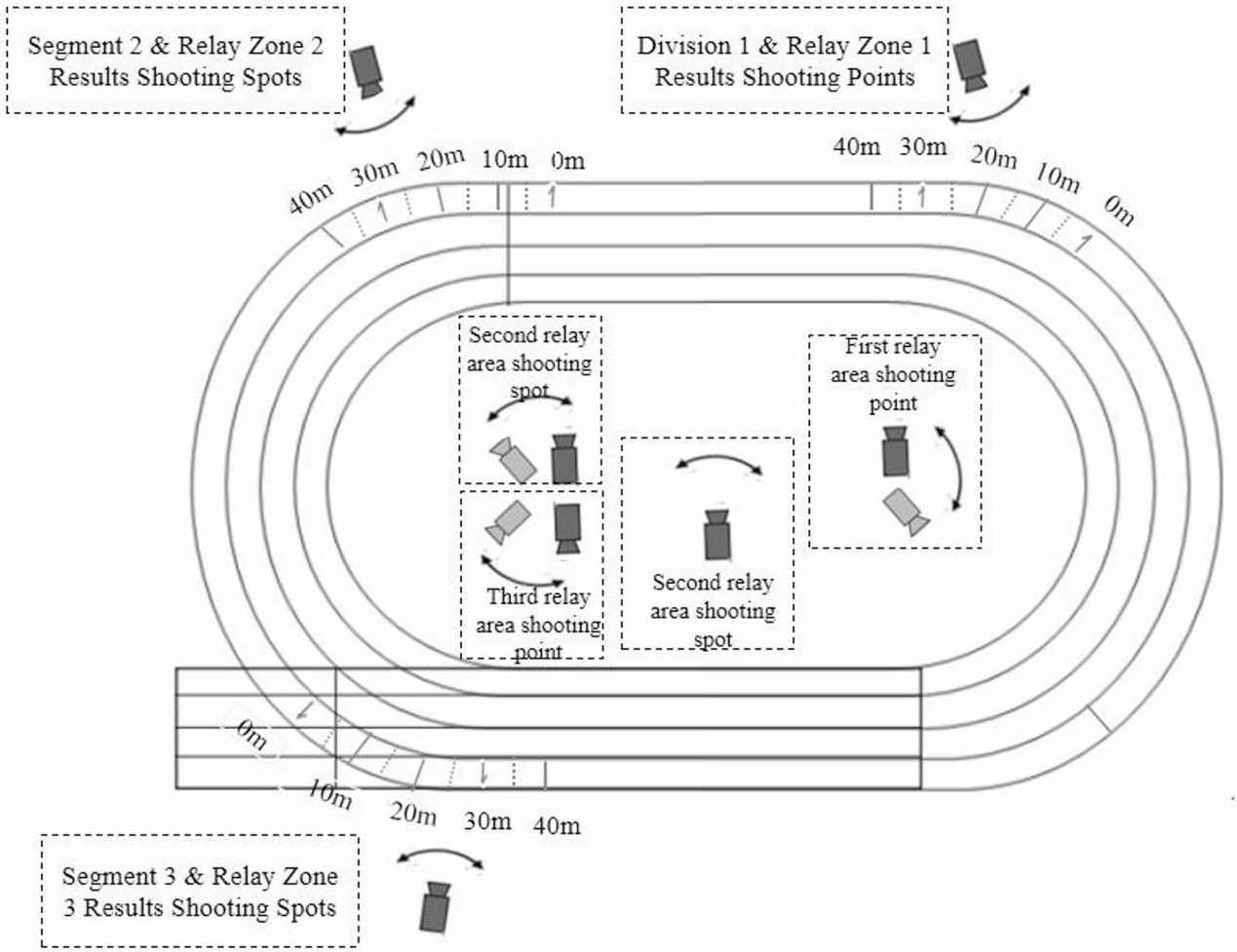

Fig. 1. Camera layout within the stadium for the relay zone training and competition.

Fig. 2. Sample video shooting.

set at 1/30. A minimum of two analysts meticulously reviewed the data points extracted from the video during the parsing process, ensuring that the error remained within a threshold of 0.004 s.

Statistical analysis

Normality was assessed using the Shapiro–Wilk test in conjunction with visual inspection of histograms. Pearson correlation analysis was carried out using GraphPad Prism9 on the relay zone time and the speed, handover time, handover position, and handover distance of the athletes, and the statistical significance level for normality testing and correlation analysis was set at $P \leq 0.05$, with "*" and "**" indicating significant and highly significant differences, respectively. The regression equation (Forward Selection) was constructed using the SPSS 27.0 statistical package (v.27, SPSS., Inc. Chicago, IL, USA). The statistical data for all indicators are presented as the mean ± standard deviation, providing a comprehensive overview of the variability and central tendency of the measurements.

Results

All data followed a normal distribution ($P > 0.05$). The relationship between relay zone 30 m time and all kinematic variables is illustrated in Table 2, highlighting the significant correlations that underpin the analysis.

During the relay zone training, the velocity of the delivery leg when entering the relay zone exhibited a strong correlation with the relay zone 30 m time ($r = -0.600$, $P < 0.01$, Fig. 3A), whereas, during the baton handover between the delivery leg and the receiving leg within the relay zone, no significant correlation was observed between the velocity of the delivery leg when handover and the relay zone 30 m time ($r = -0.341$, $P > 0.05$, Fig. 3B), the velocity of the receiving leg at 5 m was not significantly correlated with the relay zone 30 m time ($r = -0.119$, $P > 0.05$, Fig. 3C), the velocity of the receiving leg at 10 m demonstrated a significant correlation with the relay zone 30 m time ($r = -0.411$, $P < 0.05$, Fig. 3D), the velocity of the receiving leg at 15 m was significantly correlated with the relay zone 30 m time ($r = -0.474$, $P < 0.01$, Fig. 3E), the velocity of the receiving leg at the start of the baton-handover exhibited a strong correlation with the relay zone 30 m time ($r = -0.502$, $P < 0.01$, Fig. 3F).

A highly significant correlation was observed between the handover time and the relay zone 30 m time ($r = 0.389$, $P < 0.05$, Fig. 4A), whereas no significant correlation was found between the handover position and the relay zone 30 m time ($r = -0.295$, $P > 0.05$, Fig. 4B), Additionally, no significant correlation was observed between the free distance and the relay zone 30 m time ($r = -0.304$, $P > 0.05$, Fig. 4C).

In relation to the optimal technical indicators, the speed profiles of both the baton delivery and the receiving leg within the relay zone are illustrated in Fig. 5.

A series of multiple stepwise regression equations were developed by integrating the significant indicators, including X_1 = Velocity of the delivery leg when entering the relay zone, X_2 = Velocity of the receiving leg at 10 m, X_3 = Velocity of the receiving leg at 15 m, X_4 = Velocity of the receiving leg at the start of the baton-handover, X_5 = Handover time, and Y = Relay zone 30 m time (Table 3). Model 2 presents the final fitted results, indicating that the velocity of the delivery leg when entering the relay zone is t = − 4.100 ($P < 0.001$), and velocity of the receiving leg at the start of the baton-handover is t = − 3.457 ($P = 0.002$). These findings demonstrate that the partial regression coefficients for each of the variables are statistically significant. The regression equation is expressed as $Y = 5.146 - 0.121 \times_1 - 0.113 \times_4$ (adjusted $R^2 = 0.55$, $F = 15.539$, $P < 0.01$). Figure 6 illustrates the assessment of the normality of the distribution of residuals. The scatter is predominantly uniformly distributed around the reference straight line, suggesting that the data aligns well with the assumptions of the linear regression model.

Discussion

The findings of this study yield critical insights into the kinematic factors that influence relay zone performance in the men's 4 × 100 m relay. Through the application of multiple stepwise regression analyses, we established a robust relationship between the velocities of both the delivery leg and the receiving leg during the critical phases of baton-handover. Specifically, the velocity of the delivery leg when entering the relay zone and the velocity of the receiving leg at the start of the baton-handover emerged as crucial indicators of overall performance. The results underscore the significance of optimizing these velocities to enhance relay efficiency, thereby contributing to improved competitive outcomes. Furthermore, the results showed that handover time was significantly

Variables	Mean ± SD	Correlation
Velocity of the delivery leg when entering the relay zone (m/s)	10.58 ± 0.27	− 0.600**
Velocity of the delivery leg when handover (m/s)	9.88 ± 0.23	− 0.341
Velocity of the receiving leg at 5 m (m/s)	6.28 ± 0.26	− 0.119
Velocity of the receiving leg at 10 m (m/s)	7.68 ± 0.19	− 0.411*
Velocity of the receiving leg at 15 m (m/s)	8.69 ± 0.14	− 0.474**
Velocity of the receiving leg at the start of the baton-handover (m/s)	9.16 ± 0.24	− 0.502**
Handover time (s)	0.29 ± 0.11	0.389*
Handover position (m)	19.29 ± 3.20	− 0.295
Free distances (m)	1.36 ± 0.17	− 0.304

Table 2. Relationships between kinematics and relay zone 30 m time. $n = 29$; Velocity of the delivery leg when entering the relay zone, $n = 25$.

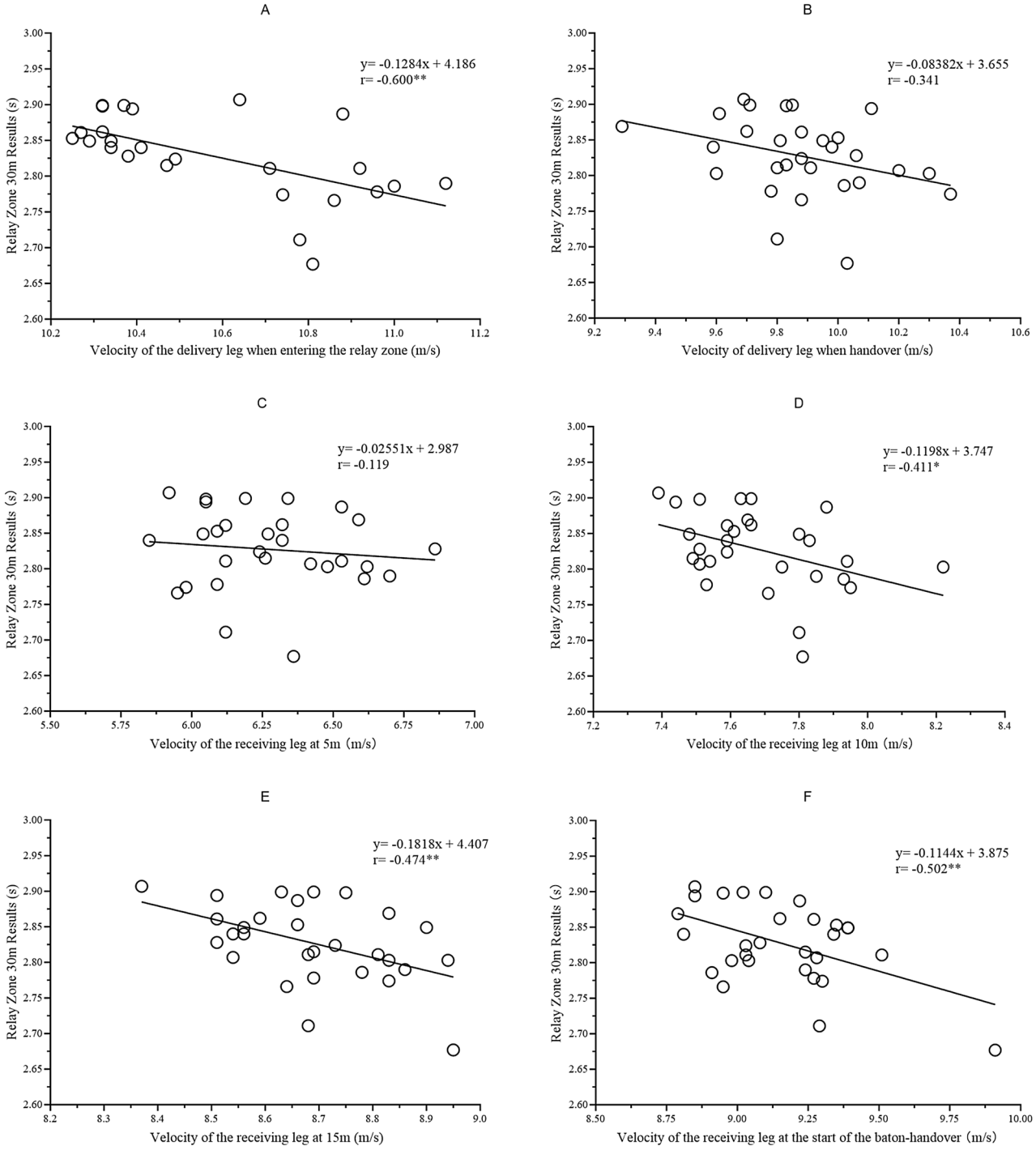

Fig. 3. Correlation between relay zone performance and speed ability.

correlated with 30 m relay zone performance. Therefore, athletes' speed capabilities and the execution quality of baton-handover technique are key factors influencing relay zone performance. These findings not only enhance our understanding of the biomechanics involved in relay events but also provide practical recommendations for training regimens designed to maximize performance in elite sprint relay competitions.

Our research has shown that there is a significant proportional relationship between the velocity of the delivery leg when entering the relay zone and relay zone 30 m time. In 4 × 100 m relay event, the delivery leg accelerates 35–40 m, reaches maximum speed at about 50–70 m, maintaining it for 20 m, and speed decreases[17,18]. For 2_{nd} and 3_{rd} leg, the longest distance completed was 130 m in competition. Hence, as the delivery leg enters the relay zone, they are in a reduced speed phase. The magnitude of their deceleration mainly depends on the ability of anaerobic glycolysis to supply energy, which is also called speed endurance, the stronger the ability, the smaller the magnitude of the deceleration at the latter stage[19,20]. Therefore, the 2_{nd} and 3_{nd} leg should have good speed

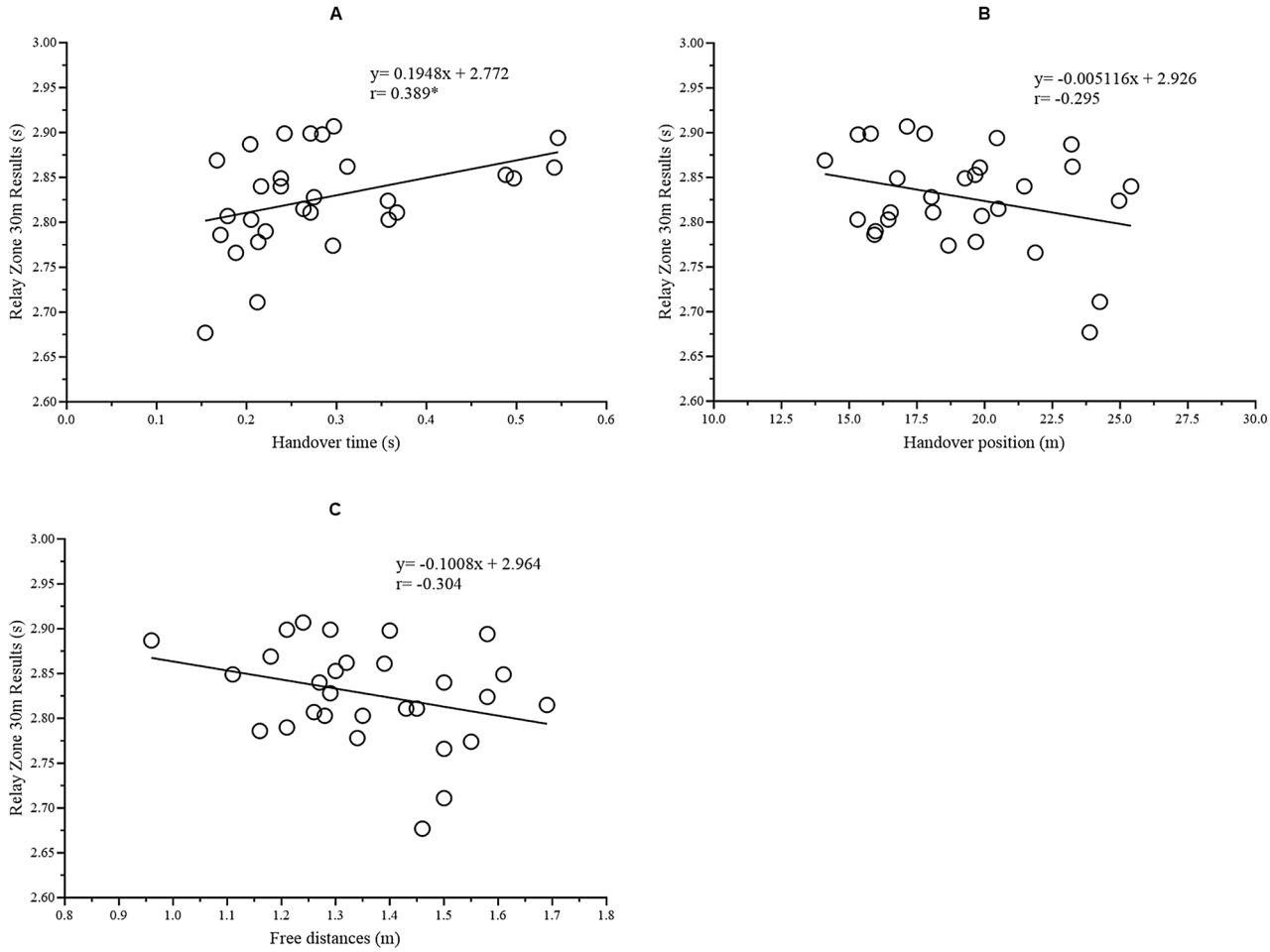

Fig. 4. Correlation between the Relay zone performance and technical parameters.

endurance, which can ensure greater instantaneous speed when entering the relay zone and avoid affecting the relay zone performance due to excessive speed reduction.

This study showed that the velocity of the receiving leg at 5 m in the relay zone is not a key factor affecting the relay zone 30 m time. However, this finding is contrary to the previous study that good acceleration is important for receiving legs to improve their baton handover speed[16]. The reason for the inconsistency of the result may be that the researcher focused on the whole acceleration phase and did not analyze the speed of the relay section on the one hand, and on the other hand, they thought that the faster speed at the start would lead to the faster speed at the time of the handover. They may also not have considered the effect of the pacing on the whole distance into consideration. In the first 5 m of the relay zone, the main task of the receiving leg is to quickly get out of a static state and gain more acceleration. At running speeds below 8 m/s, increasing stride length is more beneficial than increasing stride frequency for the athlete to gain acceleration of the body[21]. Moreover, the energy expenditure of muscle contraction in periodic exercise is squarely proportional to the rate of muscle contraction[22]. Therefore, it is more economical to increase speed during the 0–5 m acceleration phase by increasing stride length than by increasing stride frequency in relay zone, which is also important for maintaining maximal speed at the back-end deceleration phase.

When the receiving leg reaches the 10 m position in the relay zone, the speed begins to correlate significantly with the relay zone 30 m time. When the receiving leg reaches the 15 m position, the speed of the delivery leg is very significantly correlated with the relay zone 30 m time. There is an increasing trend in the significance of speed and relay zone performance at both positions. This suggests that the speed of the receiving leg is having an increasing impact on relay zone performance. As the receiving leg reaches 7.68 m/s and 8.69 m/s in these two positions of this study, it indicates that the increase in speed should be converted from stride length- to stride frequency-based at this phase.

Another observation was that the velocity of the receiving leg at the start of the baton-handover is a key factor affecting the relay zone 30 m time, in agreement with the findings of Takeshi Matsubayashi et al.[8] In the actual training and competition process, the speed of the delivery leg before the handover in the relay zone is usually greater than the speed of the receiving leg (the receiving leg reaches the maximum speed at 50–70 m). At this point, the delivery leg needs to adjust his speed according to the speed of the receiving leg and the distance in order to complete the handover. If the speed of the receiving leg is lower, the delivery leg needs to reduce his

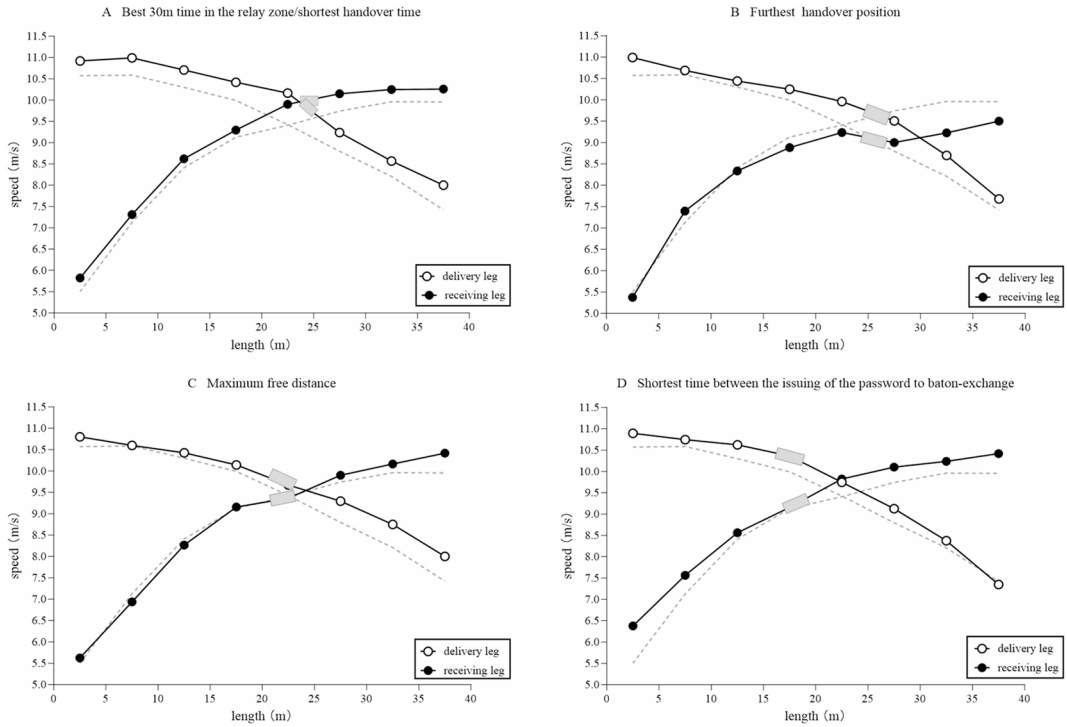

Fig. 5. Velocity profiles of delivery and receiving leg in the relay zone under conditions of optimal values of handover technique.

Model		*B*	*SE*	*β*	*t*	*P*	95% CI for *B*		Adjusted R^2
							Low limit	Upper limit	
1	Constant	4.186	0.377		11.096	< 0.001	3.405	4.966	0.33
	X_1	− 0.128	0.036	− 0.600	− 3.599	0.002	− 0.202	− 0.055	
2	Constant	5.146	0.417		12.353	< 0.001	4.282	6.010	0.55
	X_1	− 0.121	0.029	− 0.564	− 4.100	< 0.001	− 0.182	− 0.060	
	X_4	− 0.113	0.033	− 0.476	− 3.457	0.002	− 0.182	− 0.045	

Table 3. Coefficient a of multiple regression. Dependent variable: Y = Relay zone 30 m time; Independent Variable: X_1 = Velocity of the delivery leg when entering the relay zone; X_4 = Velocity of the receiving leg at the start of the baton-handover.

speed in advance to complete the handover, then the speed of the delivery leg and the receiving leg are also both very low speed in this baton exchange. On the contrary, if the speed of the receiving leg is very large, the delivery leg must maintain the maximum speed or reduce the speed by a small margin to complete the handover, which leads to a better performance in the relay zone. This is a two-sided effect at the same time superposition of the problem. Therefore, in training and competition, the receiving leg should be encouraged to increase the speed at the baton-exchange as much as possible, and the greater the speed of the receiving leg in the relay zone, which is significant for improving the relay zone performance.

Interestingly, a previous study showed that there is no significant correlation between the handover time and the 40 m performance (30 m relay zone + 10 m)[8]. However, the results of this study showed a significant correlation between athletes' handover time and the relay zone 30 m time, as shown in Fig. 4A. In agreement with the faster the receiving leg completes the handover process within the zone, the lower the loss of speed is[23]. Supporting the results of this study is also a kinematic report from the men's 4 × 100 m relay final at the 2017 World Athletics Championships, which indicated that the shortest handover time of the British team was a significant influencing factor in the winning of this competition[24]. The longer handover time resulted in slower performance in the relay zone may be due to the following reasons: first, the longer handover time represents the longer time to raise the hands, which would limit arm swing and restrict acceleration; second, the longer handover time may be caused by the inaccuracy of the handover distance or the handover position, and the athlete needs to constantly adjust the position of the body's center of gravity in order to achieve the accuracy of the handover, which affects the body's forwardness of the center of gravity speed. Therefore, the handover time should be shortened as much as possible to improve the performance of the relay zone.

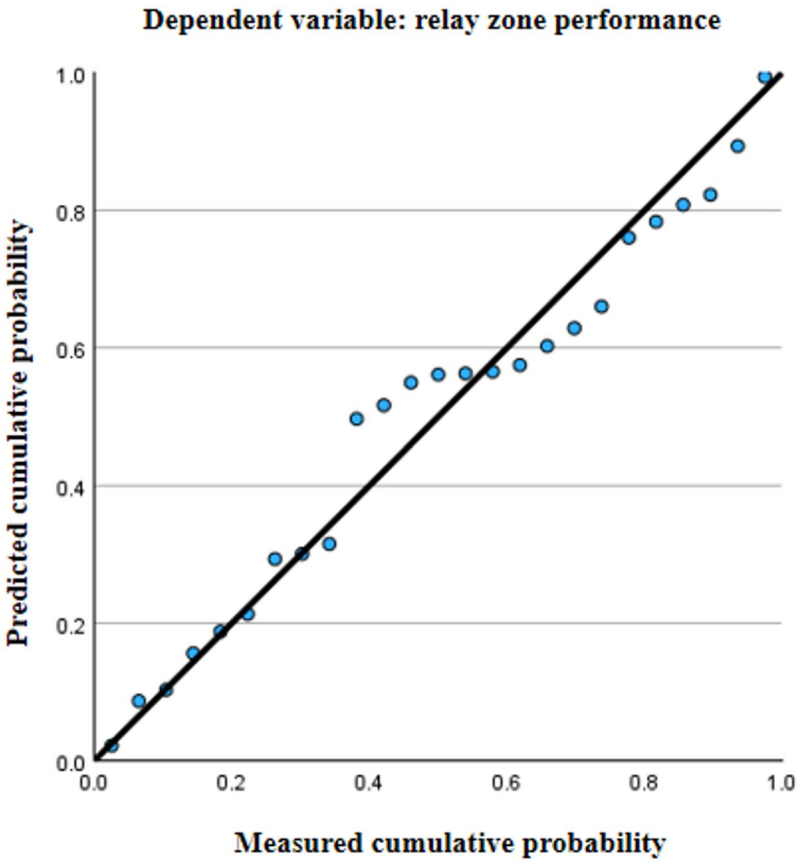

Fig. 6. Normal P-P plot of regression standardized residuals.

It should be noted that most studies showed that athletes with the baton exchange position in the final 10 m of the relay zone could obtain a better performance in the relay zone, and after the change in the rule, some studies similarly supported this conclusion[1,4,8]. However, the results of this study showed that there was no significant correlation between the handover position and the relay zone 30 m time. Theoretically, in the relay zone, the farther the handover position is, the greater the velocity during the handover. In fact, when the athletes in the relay zone 22–25 m have not yet appeared in the handover, the receiving leg usually take the deceleration in order to wait for the handover, such as case shown in Fig. 4B. At this time, although the handover in the farther position, due to the receiving leg initiative to deceleration, which is the main reason for the impact on the performance of the relay zone.

The results of this study showed that the free distance is not a significant indicator that affects the time of the relay zone. On the contrary, previous studies showed that when the free distance was increased from 0.5 m to 1.5 m, the total free distance of the three baton exchange was increased by 3 m, and the relay performance could be improved by 0.30s[1,5]. According to the regression equation for relay zone 30 m time in this study, an increase in the athletes' free distance can improve the relay zone 30 m performance. This result is similar to the results of previous studies on the quantitative relationship between the free distance and the performance in the relay zone[1,5,8]. The free distance is the distance between the delivery leg and the receiving leg during the baton handover, which is equivalent to the distance that the athletes do not need to run. During the training practice, when the athletes completed the handover before 20 m in the relay zone, the gain effect produced by the free distance may be more significant for the performance of 20 m due to the relatively shorter distance[8]. However, when the athlete completes the 30 m relay zone or 40 m distance, the smaller free distance change may not be enough to cause a significant change in the 30–40 m performance. When the free distance between the delivery leg and the receiving leg is large, may increase the difficulty of the handover between the two athletes at the same time. The resulting prolonged backward extension of the arms of the receiving leg also reduces forward velocity.

To the best of our knowledge, there are no studies on regression analysis of relay zone performance, and the multivariate linear model constructed in this study explains 55% of the relay zone performance. It is well known that there are many factors may affect relay zone performance in athletic training practice in addition to the indicators included in this study, such as the impact of the handover process on speed, the effect on speed of the receiving leg reaching backward before the handover, etc. Thus, as of now, we believe that the R^2 is an important guide to training practice in this equation. The importance of velocity of the delivery leg when entering the relay zone is reasonable because it is similar to the start velocity of 30 m. Previous study has also confirmed that horizontal start velocity can significantly affect 30 m performance[25]. Block velocity is also considered to be an indicator that accurately differentiates between different levels of 100 m athletes[26]. Velocity of the receiving leg at

the start of the baton-handover is one speed indicator that is part of the acceleration process. In sprinting, higher level athletes generate larger speed in the acceleration phase compared to lower level athletes[27]. Therefore, these two metrics are key factors in relay zone performance.

This study exhibits several strengths that significantly enhance the understanding of relay zone performance in the men's 4×100 m relay. Firstly, the application of multiple stepwise regression analyses facilitates a comprehensive examination of the relationships between various kinematic factors and performance outcomes, yielding robust and reliable results. Furthermore, the emphasis on high-level sprinters preparing for significant competitions, such as the Hangzhou Asian Games and the Paris Olympic Games, augments the relevance and applicability of the findings within elite athletic contexts. The meticulous analysis of baton exchange dynamics within the relay zones provides valuable insights that can effectively inform training practices.

Nonetheless, this study is not devoid of limitations. First, the sample size, comprising merely six high-level sprinters, may restrict the generalizability of the findings to a wider population of athletes. Second, the research is exclusively based on pre-training data, which may not comprehensively capture the effects of competition-specific variables that could potentially influence performance. Third, although the Kinovea software (version 2023.1.2; https://www.kinovea.org/) assessment method is practical, its accuracy is still lower than that of go ld-standard motion capture systems. To minimize these limitations in future research, we plan to: (i) recruit a larger and more diverse cohort of athletes to enhance statistical power and generalizability; (ii) incorporate both training and in-competition data to better reflect performance under real competitive conditions; and (iii) improve the accuracy of kinematic measurements by integrating high-speed, multi-camera motion capture systems. These refinements will help to further elucidate the complex dynamics of relay events and improve the applicability of findings to training methodologies.

Conclusion

The findings of this study underscore the critical importance of both the velocity of the delivery leg when entering the relay zone and the velocity of the receiving leg at the start of the baton-handover in enhancing overall relay performance. While the parameters of free distance and handover position were found to exert minimal influence on relay zone performance, it is evident that a reduction in handover time is significantly beneficial for optimizing performance outcomes.

Practical application

To enhance training practices for the 4×100 m relay, coaches can implement targeted methods aimed at specifically developing the speed endurance of the delivery leg. In addition, the receiving leg should be trained to apply effective pacing strategies to reach maximum velocity during the handover phase. Incorporating the "in-situ jogging–accelerating" technique into baton exchange drills can further improve the precision of hand movements and effectively reduce handover time, thereby increasing relay zone efficiency and overall competitive performance.

Data availability

Data is available from the corresponding author.

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Declarations

Competing interests

The authors declare no competing interests.

Informed consent

All participants signed informed consent forms prior to the study.

Consent for publication

This study declares that all subjects provided informed consent for publication of identifiable information/images in an online open-access publication.

Additional information

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